

Article

Ex Vivo Optical Properties Estimation for Reliable Tissue Characterization

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Abstract: Lasers are demonstrating high impact in many medical and biological applications. They have different interaction mechanisms within tissues depending on operational parameters, particularly the wavelength. In addition, the optical properties of the examined tissue (i.e., absorption and scattering properties) influence the efficacy of the applied laser. The development of optical biomedical techniques relies on the examination of tissues' optical properties, which describe the viability of tissue optical evaluation and the effect of light on the tissue. Understanding the optical properties of tissues is necessary for the interpretation and evaluation of diagnostic data, as well as the prediction of light and energy absorption for therapeutic and surgical applications. Moreover, the accuracy of many applications, including tissue removal and coagulation, depends on the tissues' spectroscopic characteristics. In the current paper, a set of ex vivo absorption and scattering coefficients of different types of biological samples (skin, skull, liver and muscle) at 650 nm laser irradiation were retrieved using an integrating sphere system paired with the Kubelka–Munk model. The obtained optical parameters were utilized to acquire the local fluence rate within the irradiated tissues based on the Monte Carlo simulation method and the diffusion approximation of the radiative transfer equation. The obtained results reveal that the optical absorption and scattering coefficients control the light propagation and distribution within biological tissues. Such an understanding refers to system design optimization, light delivery accuracy and the minimization of undesirable physiological effects such as phototoxicity or photobleaching.

Keywords: tissue optical properties; integrating sphere; Kubelka–Munk; light diffusion



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1. Introduction

Tissue optical characterization is essential for developing many optical diagnostics and therapeutic procedures, such as photobiomodulation [1], biostimulation [2], photodynamic therapy [3], laser eye surgery [4] and low-level laser therapy [5]. The energy distribution of the applied light is governed by tissue absorption and scattering characteristics [6]. Light absorption in a tissue sample allows quantitative identification of the molecules present in that sample, their concentration and their local environment, while the scattering of light provides information about micrometer-size objects (e.g., molecular weight) that cause light scattering through the sample [7]. Fundamentally, the absorption and scattering properties are not constant but vary according to the tissue's type and structure, in addition to the utilized laser's wavelength and power [8,9]. Therefore, they can be utilized for the detection and localization of tissue abnormalities, including tumors [10–12].

Due to the high dependency of light-based medical practices' outcomes on the light propagation in tissues, many studies have been proposed to estimate either the ex vivo or in vivo optical parameters of different biological tissues and organs over a wide range of light wavelengths, especially at the red and near-infrared spectral regions [13–15]. Light in this range can penetrate deep into biological tissues, and hence improve tissue monitoring

and imaging [16]. Tissue absorption and scattering coefficients are extracted indirectly using the experimental measurements of light diffuse reflectance and transmittance [17]. Such measurements are obtained via either integrating-sphere-based [18,19] or distant-detector-based systems [20,21]. An inverse mathematical model is then used to retrieve the optical coefficients from the diffuse light measurements [22].

Determining the distribution of the fluence rate inside the inspected tissue requires knowledge of the optical coefficients (absorption coefficient μ_a , scattering coefficient μ_s and anisotropy factor g) which are used as input parameters for the radiative transfer equation (RTE) [23]. Because of the complexity of accurately applying the RTE to describe light propagation in turbid biological tissues of dense scattering properties, the diffusion approximation for the RTE can be implemented [24].

Recently, lasers are demonstrating great impact in various medical fields, especially in ophthalmology [25,26], surgery [27] and dermatology [28,29]. Visible-range lasers (particularly red lasers with wavelength $\sim 600\text{--}660\text{ nm}$) have been widely used in different optical imaging modalities [30], such as diffuse optical topography/tomography [31], optical coherence tomography [32] and spatial frequency domain imaging [33,34]. Moreover, lasers at these particular wavelengths are efficiently utilized in some clinical procedures, including photodynamic therapy [35] and photobiomodulation, for the treatment of various neurological diseases [36]. Additionally, studying the variations in tissue optical properties as a function of laser beam characteristics is crucial when using low-level laser beams (i.e., between 50 and 500 mW) in therapeutic applications in order to maximize therapeutic results [37].

By distinguishing tissue scattering from tissue absorption, water, lipid and oxy- and deoxy-hemoglobin concentrations in tissue can be directly estimated using these components' well-known spectra in the inverse analysis. Moreover, the scattering coefficient can provide information about cells, cell nuclei, cell organelles and surrounding fluids. Using optogenetics as an example, providing enough light to the targeted brain objects is critical for successful optogenetic activation. The optical characteristics of the tissue, the kind of opsin expressed in the tissue, the wavelength of the light and the physical size of the targeted location all contribute to determining the amount of light necessary to excite each brain component. As a result, extensive knowledge of the optical properties of brain tissue, as well as a mathematical model that takes into account all influencing factors, are required to develop an accurate light delivery system for optogenetics.

However, controlling the light propagation and penetration within the examined tissue requires accurate and precise determination of its optical absorption and scattering parameters. Therefore, the objective of the present study was to extract the optical properties of different tissue types (ex vivo) and investigate the behavior of the light propagation and penetration accordingly. For a reliable demonstration, a variety of tissue types, including soft tissues (such as skin and liver), hard tissues (such as skull) and red muscles were studied. Tissue reflectance/transmittance was measured using a single integrating sphere and a spectrometer-based optical system. The obtained measurements were utilized in the Kubelka–Munk model for absorption and scattering coefficient calculation. The local fluence rate was obtained and visualized using the Monte Carlo simulation method and the finite-element solution of the diffusion equation.

2. Materials and Methods

2.1. Sample Preparation

Freshly excised biological samples representing soft and hard tissues were used in the present study (skin, skull, liver and muscle). Such samples were collected from chickens, rabbits and bovines. The samples were purchased from different local butcher shops and the experimental measurements were performed within less than two hours from animal slaughtering. A total of 50 samples were examined, consisting of 10 chicken breast skin, 10 rabbit skin, 10 rabbit skull, 10 chicken liver and 10 bovine muscle samples. The dimensions of the samples were $2 \times 2\text{ cm}^2$ and the thickness of the studied samples was

2 ± 0.5 mm (measured by digital micrometer, Digimatic micrometer, Mitutoyo Corporation, Sakado, Japan). Before the experimental tests, the samples were washed under running water and dried with paper towels. The relative change in the mass of the samples was observed at room temperature using a precision laboratory balance (Precisa 205 ASCS, Precisa Instruments AG, Dietikon, Switzerland) during the experiments, revealing essentially no significant change in the water content at the end of the experiment. It is worth mentioning that all experimental procedures followed the international animal care guidelines and were approved by Cairo University's institutional animal care and use committee.

2.2. Tissue Diffuse Reflectance and Transmittance Measurements

Tissue reflectance and transmittance required for reconstructing its optical scattering and absorption parameters was measured using an optical setup consisting of a single integrating sphere integrated with a digital-fiber spectrometer [19,38]. The integrating sphere is a hollow sphere coated with highly reflected material to enable collecting tissue diffuse reflectance and total transmission from specific ports. It has different ports for light entry and detector placing [39]. Additionally, the studied tissue sample was placed in different positions for collecting reflectance and transmittance separately [21] as demonstrated in Figure 1a,b. Figure 1c demonstrates the experimental configuration for measuring the collimating transmittance T_c . The reflected and transmitted light intensities were collected using the optical fiber which was connected to the spectrometer. The spectrometer was connected to a computer for data processing and analysis.

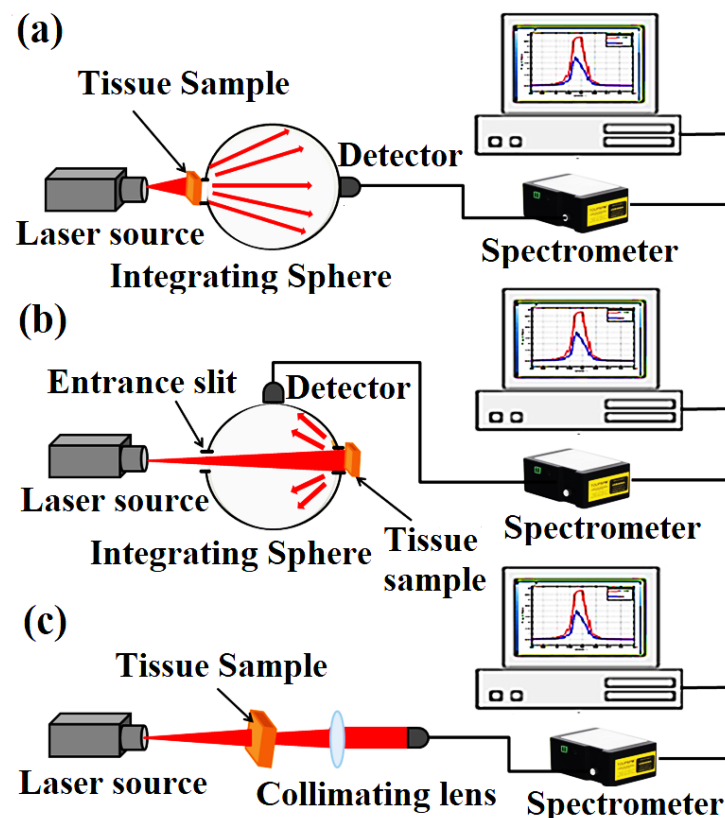


Figure 1. Experimental configurations for tissue diffusion measurement of (a) total transmittance, (b) diffuse reflectance and (c) collimating transmittance.

The numerical aperture of the light-collimating lens (shown in Figure 1c) was selected in order to guide all transmitted light towards the optical fiber that is connected to the spectrometer. All the measurements were performed using a 650 nm CW laser source (near TEM₀₀) with 130 mW incident power and 4 mm beam diameter (Model: PGL-DF,

Changchun New Industries Optoelectronics Tech. Co., Ltd., Changchun, China). The same parameters were used for the simulation.

Before placing the biological sample within the integrating sphere, the transmitted and reflected signals were checked to make sure that the light radiation was delivered inside without incurring any substantial losses. These procedures were carried out to calibrate the system. It was assumed that there would be no diffuse reflectance found in an empty integrating sphere and that the entire incident laser beam would be recorded as a transmittance. Because the sample to be measured was not present, there was no reflected intensity; instead, the laser intensity at the sphere entry was approximately equal to that transmitted to the sphere outlet. It is also worth noting that the reflectance properties of integrating sphere walls have been predicted by various studies using scalar theory, assuming that the walls completely depolarize the incident beam. These tests show that integrating spheres serve as ideal depolarizers. As a result, polarization dependence is not addressed in this study [40].

2.3. Optical Parameters Estimation

The collected diffuse measurements were used in a mathematical model for estimating the optical absorption and scattering parameters. With the combination of the Kubelka–Munk (KM) model and the Bouguer–Beer–Lambert law, the three main tissue optical parameters, the absorption coefficient μ_a , the scattering coefficient μ_s and the anisotropy g were determined. Assuming two fluxes are propagating within tissue, one propagates in the same direction as the incident light beam and the other flux propagates in the opposite direction. Two coefficients (A_{KM} and S_{KM}) are presented to demonstrate absorption and scattering of the diffuse radiation, respectively, which are related to R_d and T_d with the following equations [41,42]:

$$R_d = \frac{\sinh(S_{KM} y D)}{x \cosh(S_{KM} y D) + y \sinh(S_{KM} y D)} \quad (1)$$

$$T_d = \frac{y}{x \cosh(S_{KM} y D) + y \sinh(S_{KM} y D)} \quad (2)$$

where D is the optical thickness of the tissue sample to be considered, and the parameters x and y can be expressed in terms of:

$$x = \frac{1 + R_d^2 - T_d^2}{2R_d} \quad (3)$$

$$y = \sqrt{x^2 - 1} \quad (4)$$

The relation between S_{KM} and A_{KM} and the absorption and scattering coefficients of the sample are expressed as follows [23]:

$$A_{KM} = 2\mu_a \quad (5)$$

$$S_{KM} = \frac{3}{4}\mu_s(1 - g) - \frac{1}{4}\mu_a \quad (6)$$

The total attenuation coefficient ($\mu_t = \mu_a + \mu_s$) can be obtained using Beer's law [43]:

$$\mu_t = -\frac{1}{t} \ln T_c \quad (7)$$

where T_c is the collimated transmittance and t is the sample's thickness. The sample's thickness has a substantial impact on the experimental outcomes. The optical inhomogeneities in the sample that generate the scattering must be consistently distributed throughout and considerably smaller in size than the sample's thickness. Therefore, the thickness of the samples was chosen to allow for the necessary transmission and reflection measurements needed to apply the KM approach [44].

From KM calculations, absorption and reduced scattering coefficient ($\mu'_s = \mu_s(1 - g)$) were obtained using Equations (5) and (6). The total attenuation coefficient was determined from Beer's law, and thus the scattering coefficient can be calculated. Upon determination of (μ_a) and (μ'_s), the anisotropy factor (g) can be calculated. Finally, the optical penetration depth δ can be then calculated based on the knowledge of μ_a and μ_s using the following relation:

$$\delta = \frac{1}{\sqrt{3\mu_a(\mu_a + \mu'_s)}} \quad (8)$$

2.4. Simulating the Optical Fluence Rate Distribution

The distribution of the optical fluence at the boundary (surface) of the sample can be determined from the following diffusion equation [45]:

$$\frac{\partial \Phi(\vec{r}, t)}{\partial t} + \mu_a \Phi(\vec{r}, t) - \nabla \cdot [D \nabla \Phi(\vec{r}, t)] = S(\vec{r}, t) \quad (9)$$

where $D = \frac{1}{3(\mu_a + \mu'_s)}$ is the tissue diffusion coefficient, $\mu'_s = (1 - g)\mu_s$ is the reduced scattering coefficient and g is the anisotropy factor. $S(\vec{r}, t)$ represents the source term and $\Phi(\vec{r}, t)$ is the fluence rate. The finite-element method (FEM) was used to solve this forward diffusion equation numerically where the fluence distribution $\Phi(\vec{r}, t)$ is obtained in the domain as a function F of known optical properties $\mu(r)$ as:

$$\Phi = F(\mu) \quad (10)$$

where $\mu = [\mu_a, D]$.

In diffuse optical imaging, the image reconstruction process requires the solution of the inverse problem in which the distribution of $\Phi(\vec{r}, t)$ at the boundary is given while the optical properties of the domain are unknown; this can be represented by:

$$\mu = F^{-1}(\Phi) \quad (11)$$

Solving the inverse problem requires the minimization of the error function χ^2 , which can be calculated as:

$$\chi^2 = \frac{1}{2} \sum_{j=1}^M (\Phi_{\text{meas}} - F(\mu))^2 \quad (12)$$

where Φ_{meas} is the fluence measurements at the boundaries and $F(\mu)$ is the calculated fluence measurements ($=\Phi_{\text{calc}}$) using the forward model. In the present work, the finite-element solution of the diffusion equation was implemented to obtain fluence rate distribution images at the boundary of the sample surfaces via COMSOL Multiphysics 5.4 program software.

In COMSOL Multiphysics, the diffusion equation in the steady state (9) can be presented by the Helmholtz equation:

$$\nabla(-c \nabla u) + au = f \quad (13)$$

Identifying the parameters with Equation (9) yields:

$$u = \Phi, \quad a = \mu_a \quad (14)$$

$$c = D = \frac{1}{3(\mu_a + \mu'_s)}, \quad S = f$$

The implemented model was a rectangle of $3 \times 3 \text{ cm}^2$. A point source representing the laser source was placed at location (1.5, 1.5) to simulate the laser source position in the integrating sphere configuration, as illustrated in Figure 2.

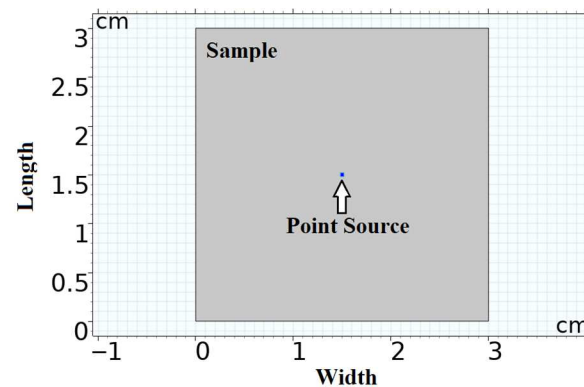


Figure 2. The implemented COMSOL model for fluence rate determination.

Additionally, the fluence distribution inside the tissue sample was simulated using the common forward numerical method named the “Monte-Carlo simulation for light propagation” [46]. In the Monte Carlo method, a pencil beam is assumed to probe the tissue sample and a number of incident photons are traced while propagating within the tissue layer(s). Each tissue layer is defined by its absorption and scattering coefficients in addition to the refractive index, anisotropy factor and thickness. Accordingly, the simulation gives information about reflectance, transmittance, absorbance and fluence [18,47].

3. Results and Discussion

From the experimentally obtained tissue diffuse reflectance and transmittance measurements, absorption and reduced scattering coefficients of the studied samples were calculated at the utilized laser wavelength (650 nm). Moreover, the obtained results were compared with previously published data from the relevant literature as summarized in Table 1. Furthermore, the resultant optical coefficients were used to calculate the optical penetration depth for each tissue as defined by Equation (8); the results are presented in Figure 3. All the measurements were performed five times and the average values with their standard deviations are presented in the table.

Table 1. The obtained tissue optical parameters compared with the related literature.

Literature	Tissue	Optical Parameters		
		μ_a, cm^{-1}	μ'_s, cm^{-1}	g
Beek et al. [48]	Rabbit skin	0.33 ± 0.02	31.6 ± 2.2	0.898 ± 0.007
Firdous et al. [49]	Chicken breast skin	2	5	0.827
Van Beers et al. [50]	Bovine muscle	0.5 ± 0.05	12 ± 3	0.91
Pehlivanöz et al. [51]	Chicken liver	2.5	6.5	--
Soleimanzad et al. [52]	Mice skull	5 ± 0.2	24 ± 1.5	--
This study	Chicken breast skin	4.91 ± 0.1	3.24 ± 2	0.81 ± 0.03
	Rabbit skin	6.04 ± 0.4	20.11 ± 2	0.85 ± 0.05
	Rabbit skull	5.91 ± 0.4	19.81 ± 1	0.90 ± 0.02
	Chicken liver	1.41 ± 0.2	8.18 ± 1	0.679 ± 0.05
	Bovine muscle	0.3 ± 0.11	10.02 ± 2	0.90 ± 0.01

In general, the proposed results show a good match in the reduced scattering coefficients values with that published in the related literature using the same wavelength, while there are some variations in the absorption coefficient values. Regarding skin samples, our obtained absorption coefficients are higher than those obtained by Beek et al. [48] and Firdous et al. [49]. Moreover, the obtained optical parameters of bovine muscle are comparable to those obtained by Van Beers et al. [50] with some variations within the acceptable range. It is worth mentioning here that the tissue preparation method and

storage criteria alter the optical parameters [14,53]. Additionally, tissue drying during the measurement procedures can also cause some alternations in the obtained values [54].

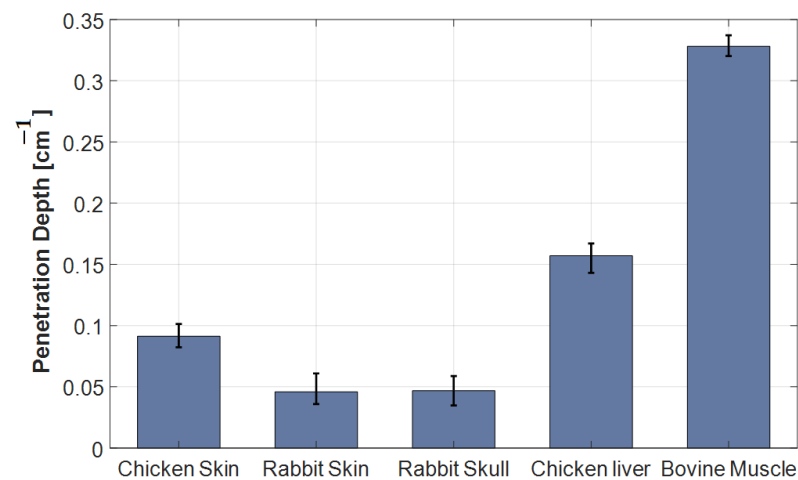


Figure 3. The calculated optical penetration depth for each studied tissue.

Using a Monte Carlo simulation (MCML), the spatial diffuse reflectance for each studied tissue was obtained as presented in Figure 4. The reflectance profile varied for each tissue due to the variation in the optical scattering and absorption characteristics.

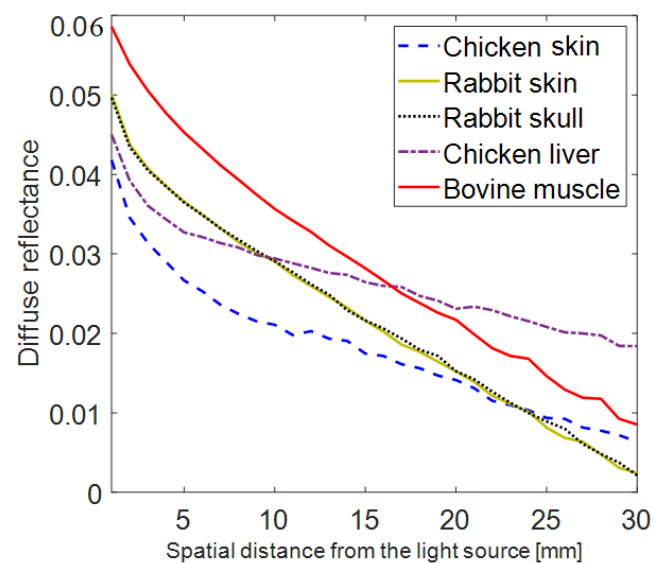


Figure 4. Spatially resolved diffuse reflectance of the studied tissues.

The distribution of the optical fluence within the target tissue samples is illustrated in Figures 5 and 6, respectively. The images were obtained by solving the diffusion equation using the finite-element method under COMSOL multiphysics platform.

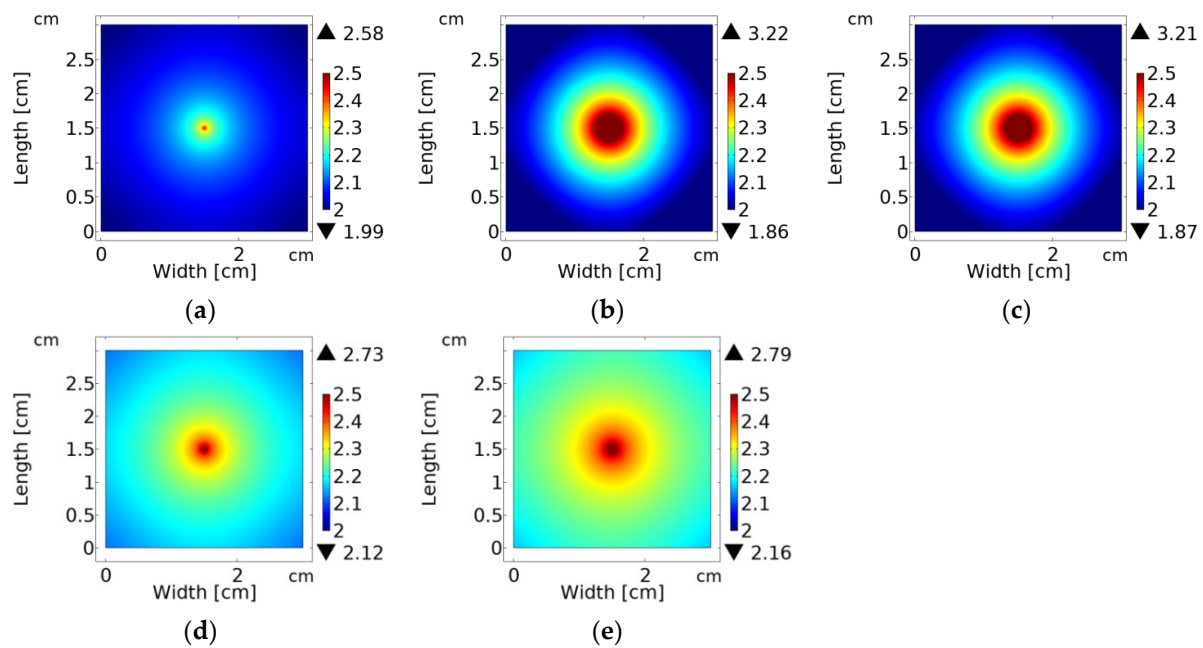


Figure 5. The optical fluence distribution at the surface of the sample: (a) chicken skin, (b) rabbit skin, (c) rabbit skull, (d) chicken liver and (e) bovine muscle (the data are presented in log scale).

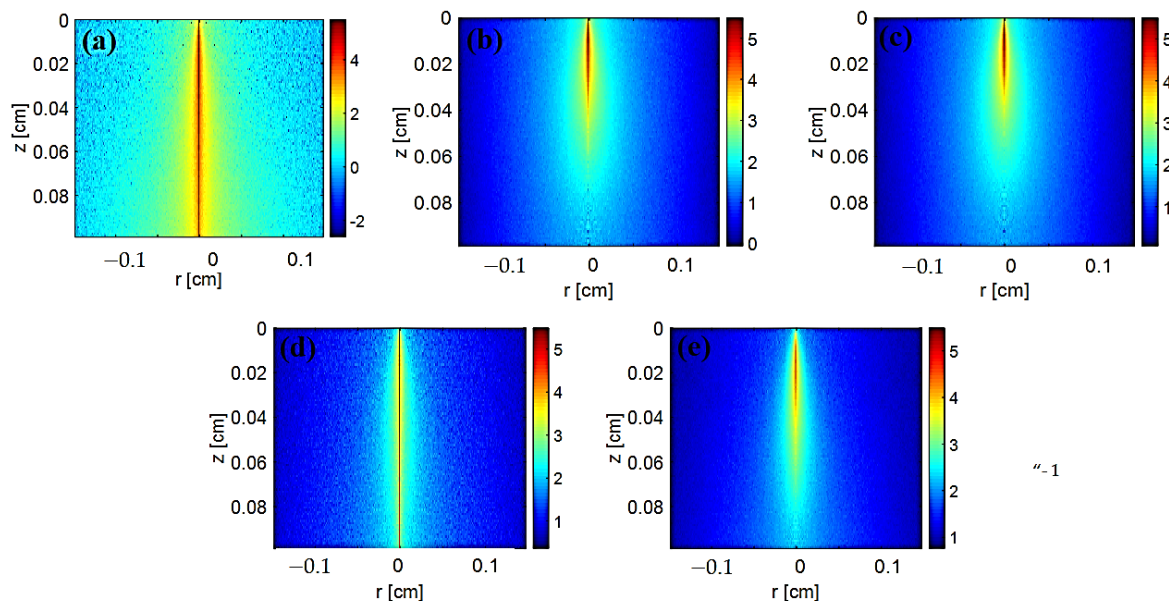


Figure 6. The optical fluence distribution with the sample depth: (a) chicken skin, (b) rabbit skin, (c) rabbit skull, (d) chicken liver and (e) bovine muscle (the data are presented in log scale).

The fluence rate at the tissue surface (Figure 5) is inversely related to the light penetration inside the tissue (Figure 6). For the higher-optical-property tissues, such as rabbit skin and skull, the distribution of the fluence rate becomes wider and more diffusive. In comparison, tissues with lower optical properties (i.e., bovine muscle, chicken skin and liver) show a more focused fluence rate and a relatively higher penetration depth. The images were obtained using a Monte Carlo simulation (MCML) with the Matlab R2018a platform.

Although precise measurements of tissue diffuse reflectance are essential for the accurate determination of the optical coefficients, tissue refractive index and anisotropy factor (g) are considered crucial elements that influence the calculation of the scattering coefficient [55,56]. It is well known that the forward scattering dominates and is connected

to the irradiation intensity in biological tissue [57], which raises the tissue anisotropy factor g [58]. Consequently, the obtained reduced scattering coefficients may decrease as a result of an increase in g . This in turn influences the diffuse reflectance and fluence distribution in tissues [59]. The relevant literature revealed that tissue diffuse reflectance increases with the scattering coefficient and decreases in tissues with high anisotropy [60,61], which agrees with our results presented in Figure 4. Moreover, a more diffusive-profile fluence distribution was evident in higher scattering samples, which also resulted in less laser penetration (see Figures 5 and 6). On the other hand, the penetration depth rises owing to an increase in the distance between the scattering events in the tissues with lower scattering.

As a future research direction, controlling the photoenergy transfer within biological tissues should be considered. When laser radiation interacts with cells, optical absorption can cause photothermal damage, which manifests as evaporation or material loss. It is critical to analyze photodamage in biological tissues subjected to laser radiation. Recently, chaotic attractors based on optical transmittance were employed to modulate light-induced alterations in thermal transfer mechanisms in human lung epithelial cancer cells [62].

4. Conclusions

We conclude that accurate estimation of light absorption and scattering coefficients is essential for many clinical and biomedical applications, such as photodynamic therapy and bio-stimulation. In such procedures, the precise modeling of light propagation is required for providing appropriate medical results. The optical absorption and scattering coefficients of various biological samples at a 650 nm laser wavelength were estimated and compared with the relevant literature in the current work. The variation in the tissue optical parameters influences the light propagation, as presented by the distribution of fluence rate through the target tissues. Wider and more diffusive fluence rate distribution was observed in the tissues with higher absorption and scattering coefficients, while focused fluence rate and higher penetration occurred in the lower-optical-coefficient tissues.

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